24BCD059

B.Tech CS Data Science (V Semester) I[st] Sessional Test 2026-2027

Time: 1 Hr
Paper Code: DSC-505

Subject: Object Oriented Programming
Max. Marks: 12

Note: Attempt All Questions.

Q1. CO1	Explain the role of different types of constructors (default, parameterized and copy constructor). Write a C++ program using a class *Student* to implement all three types of constructors and display the total number of objects created.	[4] BL-3
Q2. CO2	What is operator overloading? Write a program to overload + operator for a user defined *String* class to concatenate two strings.	[4] BL-3
Q3. CO1	a) What is an inline function? Explain its use with example.	[2] BL-2
CO2	b) What is friend function? Explain with a suitable example.	[2] BL-2